MIL-M-46336C(MU)
10 March 1965
SUPERSEDING
MIL-M-46336B(MU)
18 November 1963

MILITARY SPECIFICATION

MODULES: REPRODUCER, SIGNAL DATA: AN/GSQ-64

1. SCOPE

1.1 This specification covers five types of modules used in the Signal Data Reproducer: AN/GSQ-64. The modules covered by this specification shall be identified herein as follows:

Transistor Assembly, Power: 10531682
Amplifier Power Supply: 10531686
Amplifier, Electronic Control: 10531690
Amplifier Assembly, Photodiode: 8213319
Signal Data Reproducer Sub-Assembly: 10531698

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel: Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

FSC 1220

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DRAWINGS

U. S. Army Munitions Command

C10525749	Transistor Power 2N1542
F8213319	Amplifier Assembly, Photodiode
F8213318	Schematic, Electrical-Amplifier Assembly, Photodiode
F10531682	Transistor Assembly, Power
F10531685	Schematic, Electrical-Transistor Assembly, Po
F10531686	Amplifier, Power Supply
F10531689	Schematic, Electrical-Amplifier Power Supply
F10531690	Amplifier, Electronic Control
F10531693	Schematic, Electrical-Amplifier, Electronic Control
F10531698	Signal Data Reproducer Sub-Assembly
F10531712	Wiring Diagram, Signal Data Reproducer Sub-Assembly
F10531714	Transformer, Power, Step-down, 3 phase

PACKAGING DATA SHEETS

MIL-P-14232/P10531698	Packaging of Signal Data Reproducer Sub-Assembly: 10531698
M14002	Master Packaging Data Sheet

(Copies of specifications, standards, drawings, and publications requ' d by suppliers in connection with specific procurement functions should be tained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.2 Construction.- The modules being procured by the Government under this specification, shall be manufactured in accordance with the applicable drawing (see 2.1) and all drawings pertaining thereto.

3.3 Voltages.- Unless otherwise specified, all voltages specified in the electrical performance requirements shall be direct-current (dc).

3.4 General specification.- The contractor shall be responsible for adhere to and compliance with the following requirements of Specification MIL-F-13926. Should any conflict exist between the general specification and this specificat this specification shall take precedence.

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

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3.5 Environmental.

3.5.1 Vibration.- Each type module shall be subjected to a simple harmonic motion having an amplitude of 0.015 inch (0.03 inch maximum excursion). The frequency shall be varied continuously from 10 to 55 cycles per second and return to 10 cycles; and shall be traversed in approximately one minute. This motion shall be applied for a period of 10 minutes in each of three mutually perpendicular directions (total of 30 minutes) at standard room ambient temperature (60°F to 90°F). Subsequent to vibration, and modules shall not have been damaged or adversely affected and shall subsequently meet the electrical performance requirements of this specification.

3.5.2 Shock.- Each module shall be capable of withstanding a shock impulse of 15 g's with a duration of 11 plus or minus 1 milliseconds at standard room ambient temperature. Subsequent to shock the modules shall not have been damaged or adversely affected and shall meet the electrical performance requirements of this specification.

3.5.3 Storage temperatures.- Each module shall be capable of withstanding temperature stabilization at +160°F and at -80°F. Subsequent to such exposure the modules shall not have been damaged or adversely affected, and shall meet the electrical performance requirements of this specification at standard room ambient conditions.

3.5.4 Operating temperatures.- The modules shall meet the respective requirements of 3.6.3, 3.7.9, 3.8.11, 3.9.5 and 3.10.8 when thermally stabilized at extreme ambient temperatures of -40°F and +125°F. All other electrical performance requirements shall be met at standard room ambient temperature.

3.5.5 Humidity.- Each module shall be capable of operation in an atmosphere with a relative humidity of 90 to 95 percent. After storage for a period not in excess of four hours in an atmosphere having a relative humidity of 100 percent including condensation, the modules shall, when thoroughly dried, meet the electrical performance requirements of this specification.

3.5.6 Fungus.- Each module shall be capable of meeting the electrical and operational requirements of this specification after exposure to fungus in accordance with Specification MIL-F-13926 for a period of not less than 28 days.

3.6 Power Transistor Assembly: 10531682.- The following shall apply at standard room ambient conditions.

3.6.1 Wiring.- Wiring and connections shall be in accordance with electrical schematic, Drawing F10531685.

3.6.2 Transistors.- After final assembly of the module, all transistors shall meet the direct current gain and the D.C. leakage specified on Drawing C10525479.

3.6.3 Operating temperature extremes.- The requirements of 3.6.2 shall be met at the conditions specified in 3.5.4.

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3.7 Amplifier Power Supply: 10531686.- The circuitry shall be in accordance with electrical schematic, Drawing F10531689. Connector pins, test points, and components referenced herein are those shown on Drawing F10531689. Unless otherwise specified, the following requirements apply at the standard room ambient conditions.

3.7.1 D.C. resistance of selected circuits.

(a) The forward resistance of the bridge rectifier diodes 1N485B and 1N488 shall be between 1.5K ohms and 4K ohms.

(b) The forward resistance of the series combination CR5 and CR6 shall be between 9K ohms and 13K ohms.

(c) The resistance of the resistor-diode combinations R21 and CR30, R25 and CR37, R41 and CR41 shall be between 9.5K ohms and 11K ohms when the diodes are forward biased.

(d) The resistance across the parallel combination R13 and C7 shall be between 140 ohms and 160 ohms.

(e) R44 shall have a resistance of between 25 ohms and 29 ohms.

(f) The -150 volt (v) bridge rectifiers shall have a reverse resistance of no less than 200K ohms.

3.7.2 Reverse leakage of +25V bridge rectifier diodes.- When reverse biased by 150 V, the bridge diodes for the +25 volt supply shall have a leak of no more than 0.3 microamperes (u amps).

3.7.3 Plus 6 volt supply.- With the power transistor 2N1542 connected to the proper pins of the +6V supply, and with +25V and -20V applied to the circuit board voltage output pins, the output voltage shall be +6 ± 0.5V, with and without an 80 ohm load at the output.

3.7.4 +25 volt current limiter.- With the +6V supply circuit connections made as per 3.7.3 (no load) the following shall be exhibited:

(a) The voltage at the emitter of emitter follower Q1 shall be +24.0V ± 1.5V with respect to the +25V supply.

(b) With a 40 ohm load resistor connected between -20V and pin 4, the voltage at the emitter of Q1 shall be between -0.1V and -1V.

3.7.5 Minus 6V supply.- With -20V and ground applied to the proper pins, the voltage across the 6 volt zener, CR19 shall be 6V ± 0.3V.

3.7.6 Minus 15V and minus 12.5V supplies.- With the 2N1542 power transistor connected to the proper pins of this supply and -20V applied, the -15V output voltage shall be -15V ± 1.5V, and the -12.5V output shall be -12.5V ± 1.5V. With a 100 ohm load applied across the -15V output and a 200 ohm load applied across the -12.5V output, the voltages across these resistors shall be measured for -15V ± 1.5V and -12.5V ± 1.5V respectively.

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3.7.7 Minus 20V supply, current limiter and voltage cutout.- With the 2N1542 power transistor connected to the proper pins of this supply and -150V applied across the -150V bridge rectifier, the voltage at the -20V output, shall be -19V ± 2V. With a 36 ohm load applied at the output of the -20V, the output voltage shall be no more negative than -5V. The voltage at the base of the input emitter follower shall be between 5.7V and 6.8V of the output voltage.

3.7.8 Start trigger; flip-flop; read indicator; power transistor driver; interlock switch; stop trigger.- With the proper voltages applied, the following measurements shall be made under the stated conditions.

3.7.8.1 With a -12.5V ± 0.6 volt and ground combination input to the OR gate of the start trigger, and ground at the stop trigger input gate, the following requirements shall apply within the circuitry:

(a) The collector voltage of flip-flop transistor Q13 shall be between 0 and -0.5V.

(b) The collector voltage of flip-flop transistor Q12 shall be -9.0V ± 1.5V.

(c) The collector-emitter leakage of the read indicator transistor shall not exceed 16 u amps.

(d) The output voltage of the power transistor driver shall be -7.1V ± 0.4V.

(e) The collector-emitter leakage of the stop trigger transistor and coupling capacitor shall not exceed 20 u amps.

3.7.8.2 With the conditions of 3.7.8.1 unchanged, -12.5V ± 0.6V applied at the input of the interlock switch:

(a) The output of the power transistor driver shall drop to between 0 and -0.5V.

(b) The collector of the read indicator shall be -15V ± 1V.

(c) The voltage across R22 shall be between 5.2 and 5.3V.

3.7.8.3 With the start trigger and interlock switch inputs grounded, and -12V ± 0.6V applied to the stop trigger input, the following conditions shall exist within the circuitry:

(a) The collector of flip-flop transistor Q13 shall be more negative than -8.4V.

(b) The collector of flip-flop transistor Q12 shall be between 0 and -0.5V.

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(c) The collector voltage of the read indicator transistor sha' be between 0 and -0.5V.

(d) The output of the power transistor driver shall be between 0 and -0.5V.

(e) The collector-emitter leakage of the start trigger transistor and coupling capacitor shall not exceed 20 u amps.

3.7.9 Operating temperature extremes.- The following requirements shall be met at the conditions of 3.5.4:

3.7.9.1 D.C. resistance of selected circuits.- The requirements of 3.7.1 shall hold with exceptions in the following subparagraph measurements:

(a) Measure between 1K ohms and 4.4K ohms.

(b) Measure between 8K ohms and 15K ohms.

(c) Measure between 9.5K ohms and 15K ohms.

(d) Measure no less than 150K ohms.

3.7.9.2 Reverse leakage of +25V bridge rectifier diodes.- The requirements of 3.7.2 shall apply with the exception that the leakage shall not exceed 33 u amps.

3.7.9.3 Plus 6V supply.- The requirements of 3.7.3 shall apply.

3.7.9.4 Plus 25V current limiter.- The requirements of 3.7.4 shall apply with the exception that the emitter voltage of Q1 shall be +23.7 ± 2V (no load).

3.7.9.5 Minus 6V supply.- The requirements of 3.7.5 shall apply except the zener voltage shall be 6 ± 0.4V.

3.7.9.6 Minus 15V and minus 12.5V supplies.- The requirements of 3.7.6 shall apply.

3.7.9.7 Minus 20V supply, current limiter and voltage cutout.- The requirements of 3.7.7 shall apply.

3.7.9.8 Start trigger; flip-flop; read indicator; power transistor driver; interlock switch; stop trigger.- The requirements of 3.7.8 shall apply with exceptions to the following, as denoted by subparagraphs:

3.7.8.1(c) Leakage shall not exceed 50 u amps.
3.7.8.1(d) The output voltage shall be -7.1V ± 0.9V.
3.7.8.2(a) The voltage shall be between 0 and -1V.
3.7.8.2(b) The voltage shall be -15V ± 1.5V.
3.7.8.3(c) The voltage shall be between 0 and -1V.
3.7.8.3(d) The voltage shall be between 0 and -1V.

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3.8 Amplifier Electronic Control: 10531690.- The circuitry shall be in accordance with electrical schematic, Drawing FI0531693. The symbols referenced in subsequent paragraphs are identified functionally on that drawing. Unless otherwise specified, the following requirements apply at the standard room ambient conditions.

3.8.1 Brake-solenoid circuit.

3.8.1.1 The brake-solenoid diode shall have:

(a) A forward voltage drop no greater than 1.0V at 150 ± 7.5 milliamperes (ma).

(b) A reverse leakage of no more than 15 u amps at 150V.

3.8.1.2 The associated resistor shall have a value of 51K ohms ± 5%. The capacitor shall have a value of 13 micro farads (uf), plus 3.0, minus 1.5 microfarads.

3.8.2 Minus 3V supply.- The supply zener voltage shall be -3.5 ± 0.5V at 10 ± 1ma. The associated resistor shall be 470 ohms ± 5%.

3.8.3 Resistor check.- Resistor R31 shall be 91K ohms ± 5%.

3.8.4 Capacitance of automatic gain control networks.- The capacitance of the individual AGC circuit shall conform to their appropriate drawings.

3.8.5 Power transistor driver, solenoid switch.

(a) With the input of the driver grounded, the voltage at the output of the solenoid switch divider shall be -5.75 ± 0.4 volts.

(b) With -15 ± 1V applied to the driver input:

(1) The voltage at TP18 shall be -7.4 ± 0.5V.

(2) The voltage at the solenoid switch divider shall be more positive than +0.5V.

3.8.6 Inverter networks.

3.8.6.1 With ground at the input of inverter 1 (Q1), and -15 ± 1V thru a 1K ohm resistor to the input of inverter 2:

(a) The collector of Q1 shall be between -14.1 V and -15.6 V.

(b) The collector of Q4 shall be between 0 and -0.5 V.

(c) The voltage across the 1K ohm resistor shall be less than 0.1 V.

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3.8.6.2 With ground and -15V resistor inputs of inverters 1 and 2 interchanged:

(a) The collector Q1 shall be between 0 and -0.5V.

(b) The collector of Q4 shall be between -14.1V and -15.6V.

(c) The voltage across the 1K ohm resistor shall be less than 0.1V.

3.8.6.3 With ground at the inputs of both inverters:

(a) The collector of Q1 shall be between -14.1V and -15.6V.

(b) The collector of Q4 shall be between -14.1V and -15.6V.

3.8.6.4 With ground at the input of inverter 1, and both combinations of ground and -15 volts thru a 1K ohm resistor, at the CR7-CR8 inputs to inverter 2, the voltage across the 1K ohm resistor shall be less than 0.1V.

3.8.7 Stop switch.- With the inhibit from inverter 1 (Q1) removed, a -15 ± 1V input to the stop switch input shall produce a pulse as shown in Figure 1.

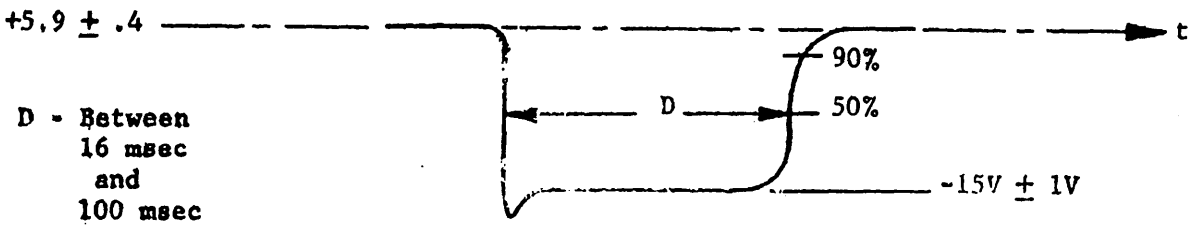


FIGURE 1

3.8.8 Delay and delay-inhibit.- With the inhibit from inverter 1 (Q1) removed, the delay and delay-inhibit networks shall cause a response at the collector of Q5 which is delayed at least 80 m sec but not greater than 400 m sec from an input of -15 ± 0.7V to the delay circuit. The response shall be a change in voltage levels, starting at -0.2 ± .2V and after the 80 m sec delay, shall go to a voltage no less negative than -13V.

3.8.9 Emitter follower.- The emitter follower and its associated circuitry shall perform properly at input voltages of ground and -15V. The emitter output shall follow within 0.5V of the base input.

3.8.10 Voltage conversion emitter followers.- With a 150 ohm load to ground on the emitter of the circuits and the clamping diode biased at -3V, an input of ground shall produce an emitter voltage between 0 and -0.5V. An input of -12.5V shall produce an emitter voltage of -2.7 ± 0.3V.

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3.8.11 Operating temperature extremes.- The following requirements shall be met at the conditions of 3.5.4.

3.8.11.1 Brake-solenoid circuit.- The brake-solenoid diode shall have:

(a) A forward voltage drop no greater than 1.5V at 150 ± 7.5 ma.

(b) A reverse leakage of no more than 50 u amps at 150V.

3.8.11.2 Minus 3V supply.- The zener voltage shall be -3.5 ± 0.5V at 10 ± 1 ma.

3.8.11.3 Power transistor driver - solenoid switch.

(a) With the input of the driver at ground, the output of the solenoid switch driver shall be -5.7 ± 0.5V.

(b) With -15V applied to the driver input the voltage at TP18 shall be -7.4 ± 0.7V, and the voltage at the solenoid switch divider shall be more positive than + 0.3V.

3.8.11.4 Inverter networks.- The requirements of 3.8.6 shall apply except in 3.8.6.1(c), 3.8.6.2(c) and 3.8.6.4, the voltage across the 1K ohm resistor shall be less than 0.3V.

3.8.11.5 Stop switch.- The requirements of 3.8.7 shall apply.

3.8.11.6 Delay and delay-inhibit.- The requirements of 3.8.8 shall apply.

3.8.11.7 Emitter follower.- The requirements of 3.8.9 shall apply.

3.8.11.8 Voltage conversion emitter followers.- The requirements of 3.8.10 shall apply except:

(a) A ground input shall produce an emitter voltage between 0 and -0.6V.

(b) A -12.5V input shall produce an emitter voltage of -2.7 ± 0.5V.

3.9 Amplifier Assembly, Photodiode: 8213319.- The circuitry shall be in accordance with electrical schematic, Drawing F8213318. The symbols referenced in subsequent paragraphs are identified in aforementioned drawing. Unless otherwise specified, the following requirements apply at the standard room ambient conditions.

3.9.1 Resistor selection.- With the input of Figure 2A applied to each of the photodiode inputs, resistors R56 thru R63 should be selected to produce an amplifier output consistent with Figure 2B.

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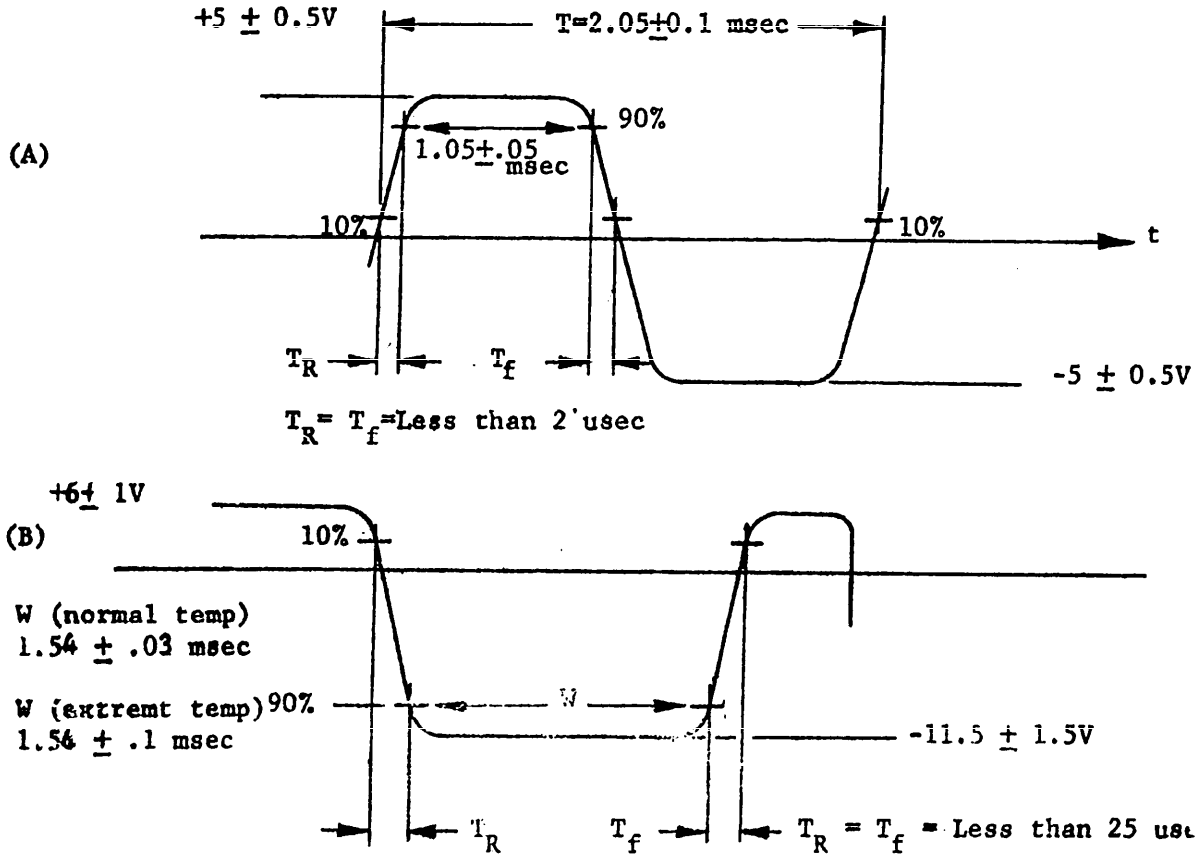


FIGURE 2

3.9.2 Photodiode amplifier inhibit diodes.- The inhibit diodes when grounded shall inhibit the output of the photodiode amplifiers.

3.9.3 TR amplifier.

(a) With a -15 ± 0.7V input, the following amplifier voltage shall be as indicated:

(1) at TP45, the voltage shall be -6.2 ± 0.4V.

(2) the output voltage shall be -12.3 ± 0.7V.

(b) With a ground input, the output voltage shall be +6 ± 0.3V.

3.9.4 Clock amplifier.- With a symmetrical 4 ± 0.2V (+2V to -2V) peak-to-peak 480 ± 20 cps square wave input to the clock emitter follower, the output of the first stage amplifier shall be as shown in Figure 3. The output of the second stage amplifier shall be of the same shape as Figure 3 except the negative limit shall be -12.5V ± 1.5V.

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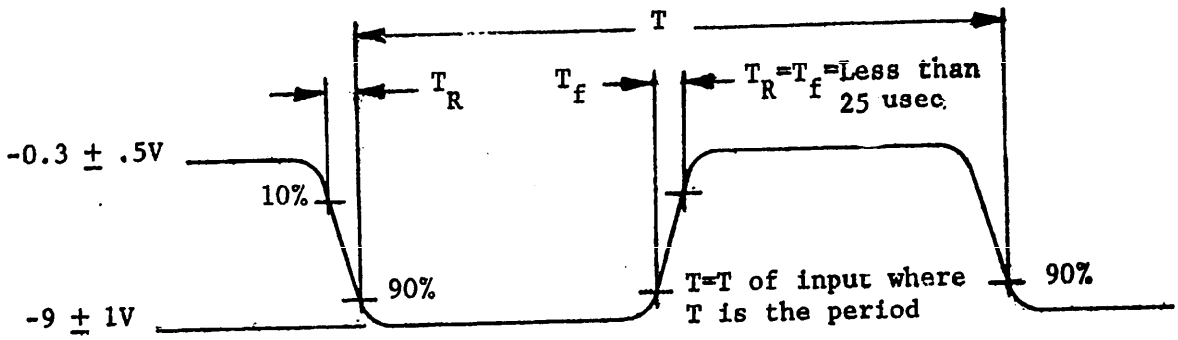


FIGURE 3

3.9.5 Operating temperature extremes.- The following requirements shall apply at the conditions of 3.5.4.

3.9.5.1 Photodiode amplifier and inhibit diode.

(a) The requirements of 3.9.1 shall apply.

(b) Application of ground to the inhibit diodes shall inhibit the output wave.

3.9.5.2 TR amplifier.-

(a) With a -15 ± 0.7V input to the amplifier:

(1) The voltage at TP45 shall be -6.2 ± 0.6V.

(2) The voltage at the amplifier output shall be -12.3 ± 1V.

(b) With a ground input, the amplifier output voltage shall be +6 ± 0.6V.

3.9.5.3 Clock amplifier.- The requirements of 3.9.4 shall apply.

3.10 Signal Data Reproducer Sub-Assembly: 10531698.- The following specifications apply at standard room ambient conditions, unless otherwise stated. Pin connectors refer to electrical schematic, Drawing F10531712.

3.10.1 Wiring.- Wiring shall be in accordance with Drawing F10531712.

3.10.2 Zener voltages.- CR8 shall have a zener voltage between -15.2 to -16.8V when drawing 0.5 amp.

3.10.3 Brake solenoid switch.- Q2 shall have a maximum collector-base cutoff current of 100 u amps with V_{CBO} = -2V. Q2 shall have a minimum D.C. current gain of 25 and a maximum of 50 with I_C = 5 amps and V_{CE} = -2V.

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3.10.4 Drive solenoid switch.- Q1 shall have a maximum collector-base cutoff current of 100 u amps with V_{CBO} = -2V. Q1 shall have a minimum D.C. current gain of 20 and a maximum of 40 with I_C = 5 amps and V_{CE} = -2V.

3.10.5 Transformer supply.- The transformer input and output voltages shall be in accordance with those specified on Drawing F10531714.

3.10.6 D.C. resistance of selected circuits.

(a) All resistor values and tolerances shall conform to those specified on Drawing F10531712.

(b) The parallel combination of CR3, CR5, and CR7 shall have a reverse impedance of not less than 50,000 ohms. The parallel combination of CR2, CR4, and CR6 shall have a reverse impedance of not less than 50,000 ohms.

3.10.7 Clock inhibit diode.- CR9 shall have a forward voltage drop of less than 0.3 volts when drawing 2 m amps. The reverse leakage current shall not exceed 50 u amps with an inverse voltage of 20V.

3.10.8 Operating temperature extremes.- The following requirements shall be met at the conditions of 3.5.4.

3.10.8.1 Zener voltages.- CR8 shall have a zener voltage between -14.4V and -17.2V when drawing a current of 0.5 amp.

3.10.8.2 Break solenoid switch.- Q2 shall have a maximum collector-base cutoff current of 15 m amps with V_{CBO} = -100V.

3.10.8,3 Drive solenoid switch.- Q1 shall have a maximum collector-base cutoff current of 15 m amps with V_{CBO} = -60V.

3.10.8.4 Transformer supply.- The requirements of 3.10.5 shall apply.

3.10.8.5 D.C. resistance of selected circuits.- The requirements of 3.10.6 shall apply.

3.10.8.6 Clock inhibit diode.- CR9 shall have a forward voltage drop of less than .5V when drawing 2 m amps. The reverse leakage current shall not exceed 100 u amps with an inverse voltage of 20V.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 General provisions.- The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Submission of product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examinations and tests.- Except where otherwise specified herein, components and sub-assemblies shall be inspected in accordance with the "Inspection provisions for components and sub-assemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section3 herein shall be performed in accordance with Tables I to XII inclusive and the test methods and procedures in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.2.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.2.4 Functional tests.- The requirements and tests in Tables I to V inclusive shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60° to 90°F).

TABLE I

Power Transistor: 10531682
(100 percent inspection)

	Characteristics	Requirements	Test Procedures
101.	Wiring	3.6.1	4.5.2.1.1
102.	D.C. Current gain	3.6.2	4.5.2.1.2
103.	D.C. leakage current	3.6.2	4.5.2.1.3

TABLE II

Amplifier-Power Supply: 10531686
(100 percent inspection)

104.	D.C. resistance of selected circuits	3.7.1	4.5.2.2.1
105.	Reverse leakage of +25V bridge rectifier diodes	3.7.2	4.5.2.2.2
106.	Plus 6 volt supply	3.7.3	4.5.2.2.3
107.	Plus 25 volt current limiter	3.7.4	4.5.2.2.4
108.	Minus 6 volt supply	3.7.5	4.5.2.2.5

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TABLE II (Cont'd)

	Characteristics	Requirements	Test Procedure
109.	Minus 15 volt and minus 12.5 volt supplies	3.7.6	4.5.2.2.6
110.	Minus 20 volt supply, current limiter and voltage cutout	3.7.7	4.5.2.2.7
111.	Start trigger, flip-flop, read indicator, power transistor drivers, interlock switch, stop trigger	3.7.8	4.5.2.2.8

TABLE III

Electronic Control Amplifier: 10531690
(100 percent inspection)

	Characteristics	Requirements	Test Procedure
112.	Brake Solenoid Circuit	3.8.1	4.5.2.3.1
113.	Minus 3 volt supply	3.8.2	4.5.2.3.2
114.	Resistor check	3.8.3	4.5.2.3.3
115.	Capacitance of AGC Networks	3.8.4	4.5.2.3.4
116.	Power transistor drivers, solenoid switch	3.8.5	4.5.2.3.5
117.	Inverter networks	3.8.6	4.5.2.3
118.	Stop switch performance	3.8.7	4.5.2.3..
119.	Delay, delay-inhibit	3.8.8	4.5.2.3.8
120.	Emitter follower	3.8.9	4.5.2.3.9
121.	Voltage conversion emitter followers	3.8.10	4.5.2.3.10

TABLE IV

Photodiode Amplifier: 8213319
(100 percent inspection)

122.	D.C. resistor selection	3.9.1	4.5.2.4.2
123.	Photodiode amplifier inhibit diode	3.9.2	4.5.2.4.3
124.	TR amplifier	3.9.3	4.5.2.4.4
125.	Clock amplifier	3.9.4	4.5.2.4.5

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output voltage

TABLE V

Signal Data Reproducer Sub-Assembly: 10531698
(100 percent inspection)

	Characteristics	Requirements	Test Procedures
126.	Wiring	3.10.1	4.5.2.5.1
127.	Zener voltages	3.10.2	4.5.2.5.2
128.	Brake solenoid switch	3.10.3	4.5.2.5.3
129.	Drive solenoid switch	3.10.4	4.5.2.5.4
130.	Transformer supply	3.10.5	4.5.2.5.5
131.	D.C. resistance of selected circuits	3.10.6	4.5.2.5.6
132.	Clock inhibit diode	3.10.7	4.5.2.5.7

4.3 Standard sampling.

4.3.1 General.- One of each type module shall be selected at ramdom by a Government representative as a standard sample from each 100 produced. The sample shall meet the requirements and tests in Table VI.

TABLE VI

	Characteristics	Requirements	Test Procedures
301.	Materials	3.1	Applicable Drawings and specifications - Visual
302.	Construction	3.2	Applicable Drawings - Visual
302.	General specifications	3.4	MIL-F-13926 - Visual

4.3.2 Environmental.- Three of each type module shall be selected at random by a Government representative as a standard sample from each 50 produced, or from each month's production, whichever occurs first. The samples shall have met the requirements and tests of their respective table for the 100% inspection required under 4.2.4 and shall then meet the requirements and tests in Tables VII through XII, as applicable.

TABLE VII

All Modules

	Characteristics	Requirements	Test Procedures
304.	Vibration	3.5.1	4.5.1.2
305.	Shock	3.5.2	4.5.1.3
306.	Humidity - non-operating	3.5.5	4.5.1.4.1
307.	Humidity - operating	3.5.5	4.5.1.4.2

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NOTE: Samples having passed the above requirements and tests shall be subjected to the storage temperature requirements and tests of 3.5 and 4.5.1. and shall then meet the requirements and tests in Tables VIII through XII, as applicable, at the extreme operating temperatures and tests required by 3.5.4 and 4.5.1.5.

TABLE VIII

Power Transistor Assembly: 10531682
(-40+ and +125°F)

	Characteristics	Requirements	Test Procedures
308.	D.C. current gain	3.6.3	4.5.2.1.4.1
309.	D.C. leakage	3.6.3	4.5.2.1.4.2

TABLE IX

Amplifier Power Supply: 10531686
(-40° and +125°F)

310.	D.C. resistance of selected circuits	3.7.9.1	4.5.2.2.1
311.	Reverse leakage of +125V diodes	3.7.9.2	4.5.2.2.2
312.	Plus 6V supply	3.7.9.3	4.5.2.2.3
313.	Plus 25V current limiter	3.7.9.4	4.5.2.2.4
314.	Minus 6V supply	3.7.9.5	4.5.2.2.5
315.	Minus 15V and minus 25V supplies	3.7.9.6	4.5.2.2.6
316.	Minus 20V supply current limiter and voltage cut-out	3.7.9.7	4.5.2.2.7
317.	Start trigger; flip-flop, read indicator; power transistor drivers, interlock switch, and stop trigger	3.7.9.8	4.5.2.2.8

TABLE X

Electronic Control Amplifier: 10531690
(-40° and +125°F)

318.	Brake solenoid circuit	3.8.11.1	4.5.2.3.11.1
319.	Minus 3V supply	3.8.11.2	4.5.2.3.11.2
320.	Power transistor driver	3.8.11.3	4.5.2.3.11.3
321.	Inverter networks	3.8.11.4	4.5.2.3.11.4
322.	Stop switch	3.8.11.5	4.5.2.3.11.5
323.	Delay and delay-inhibit	3.8.11.6	4.5.2.3.11.6
324.	Emitter follower	3.8.11.7	4.5.2.3.11.7
325.	Voltage conversion emitter followers	3.8.11.8	4.5.2.3.11.8

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and vertical - shall cover positions to determine compliance with shall up

TABLE XI

Amplifier Assembly, Photodiode: 8213319
(-40° and +125°F)

	Characteristics	Requirements	Test Procedures
326.	Photodiode amplifier inhibit diodes	3.9.5.1	4.5.2.4.6.1
327.	TR amplifier	3.9.5.2	4.5.2.4.6.2
328.	Clock amplifier	3.9.5.3	4.5.2.4.6.3

TABLE XII

Signal Data Reproducer Sub-assembly: 10531968
(-40° and +125°F)

329.	Zener voltages	3.10.8.1	4.5.2.5.8.1
330.	Brake solenoid switch	3.10.8.2	4.5.2.5.8.2
331.	Drive solenoid switch	3.10.8.3	4.5.2.5.8.3
332.	Transformer supply	3.10.8.4	4.5.2.5.8.4
333.	D.C. resistance of selected circuits	3.10.8.5	4.5.2.5.8.5
334.	Clock inhibit diode	3.10.8.6	4.5.2.5.8.6

4.3.3 Failure of sample.- Should any one standard sample of a specific type module fail to meet the specified test requirements, acceptance of the represented inspection lot for that type will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.2.3).

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

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4.4.1.1 Test equipment.- In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.5:

Item Number	Description
1. D.C. Power Supply (3)	D.C. variable, from zero to 30 volts, current requirement of at least 1 amp. The peak-to-peak ripple and noise shall be less than 200 millivolts and the output impedance shall be less than 10 ohms.
D.C. Power Supply (1)	D.C. variable, from zero to 150 volts, current requirement of at least 200 ma. The peak-to-peak ripple and noise shall be less than 400mv and the output impedance shall be less than 10 ohms.
2. Resistors:	
50,000 ohm potentiometer	1/4 watt
5 ohm ± 5%	45 watt.
10 ohm ± 5%	5 watt
14 ohm ± 5%	20 watt
15 ohm ± 5%	20 watt
40 ohm ± 5%	10 watt
65 ohm ± 5%	10 watt
80 ohm ± 5%	1 watt
100 ohm ± 5%	2 watt
150 ohm ± 5%	1/2 watt
200 ohm ± 5%	1/2 watt
400 ohm ± 5%	2 watt
1.0 kilo-ohm ± 5%	1/4 watt
1.0 kilo-ohm ± 5%	3 watt
3.6 kilo-ohm ± 5%	1/2 watt
10.0 kilo-ohm ± 5%	1/4 watt
10.0 kilo-ohm ± 5%	1/2 watt
27.0 kilo-ohm ± 5%	1/4 watt
300.0 kilo-ohm ± 5%	1/4 watt
1000.0 kilo-ohm ± 5%	1/4 watt
3. Oscilloscope:	Single trace with maximum rise time of 0.05 microseconds, minimum input impedance of 2 megohms and input capacitance less than 25 micro-farads, vertical sensitivity of 50 millivolts per centimeter, sweep rate of 75 micro-seconds per centimeter or greater.
4. Micro-ammeter:	Capable of measuring 0 to 200 micro-amperes with the accuracy specified in 4.4.2.2, and an input impedance not in excess of 250 ohms.

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Item Number	Description
5. Capacitance Bridge:	Capable of measuring 1 to 200 microfarads with the accuracy specified in 4.4.2.2.
6. Ohmmeter:	Capable of measuring 0 to 1 megohm with the accuracy specified in 4.4.2.2.
7. Voltmeter:	Capable of measuring 0 to 200 volts D.C., and 120 volts RMS with the accuracy specified in 4.4.2.2, with a D.C. input resistance of 10 megohms or greater.
8. Ammeter:	Capable of measuring 0 to 1 amp with the accuracy specified in 4.4.2.2.
9. Square Wave Generator:	Capable of generating a variable amplitude pulse of 0 to 12 volts peak-to-peak at a variable frequency of D.C. to 500 cps, with a maximum rise time of 1 micro-second.
10. Resistance Bridge:	Capable of measuring 75,000 to 110,000 ohms with an accuracy of ± 0.5%.

4.5 Test and methods and procedures.

4.5.1 Environmental.

4.5.1.1 Vibration.- The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adjusting the amplitudes and frequencies to keep them within the specified limits of 3.5.1. The measuring instruments of the testing device shall be capable of measuring the vibration amplitude within plus or minus 5 percent, and capable of measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements specified in 3.5.1 with the module rigidly mounted in its normal operating position. Subsequent to the vibration test, the unit shall show no damage or defect resulting from the test when examined by visual and tactile means. The unit shall then be inspected as specified in 4.5.1.2.

4.5.1.2 Shock.- This test shall be conducted in accordance with the conditions of 3.5.2 and Procedure II of the shock test defined in MIL-F-13926 except that impact shocks shall not exceed 5 G's. The module shall be subsequently inspected for physical damage or defects in compliance with 3.5.2. The unit shall then be inspected as specified in 4.5.1.3.

4.5.1.3 Humidity.

4.5.1.3.1 Non-operating.- This test shall be conducted in accordance with the conditions in 3.5.5 and the humidity test defined in Procedure I of MIL-F-13926 except that the 4-1/2 hour period shall be reduced to 2 hours, the 6 hour period reduced to 2 hours, and the total test period shall not exceed 5.5 hours per cycle. The unit shall then be thoroughly dried and then subjected to the test in 4.5.1.3.2.

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4.5.1.3.2 Operating.- This test shall be conducted in accordance with the conditions in 3.5.5 and the humidity test defined in Procedure III of MIL-F-13936 except that the humidity shall be maintained at 95 ± 5 percent for the 24 hour period, and the test shall cease at the end of that period (Method 106 of MIL-STD-202 shall be excluded). Throughout the test the unit shall be operated in accordance with 4.5.2.1.4, 4.5.2.2.9, 4.5.2.3.11, 4.5.2.4.6 and 4.5.2.5.8 and shall meet the respective requirements referenced therein.

4.5.1.4 Temperature (operating and storage).- After being subjected to all environmental tests specified above, the module shall be subjected to Test Procedure I specified in MIL-F-13926, except the temperatures specified in 3.5.3 and 3.5.4 shall apply. While at the operating temperatures of 3.5.4 the unit shall be tested in accordance with 4.5.2.1.4, 4.5.2.2.9, 4.5.2.3.11, 4.5.2.4.6 and 4.5.2.5.8 to determine compliance with the requirements of 3.5.4. The temperature shall then be returned to standard temperature (60° to 90°F) and the unit shall be tested in accordance with 4.5.2 (between 60° to 90°F only) to determine compliance with 3.5.3.

4.5.1.5 Fungus.- To determine compliance with 3.5.6, one of the first 5 of each type module produced shall be submitted for testing in accordance with the fungus test defined in MIL-F-13926. After approval, subsequent acceptance for fungus requirements may be by certification.

4.5.2 Performance.

4.5.2.1 Power Transistor Assembly: 10531682.- The following tests shall be conducted at the standard room ambient temperature (60° to 90°F) unless otherwise specified, to determine compliance with 3.6. Pin connectors refer to electrical schematic F10531685.

4.5.2.1.1 Wiring.- Wiring and connections shall be visually inspected and shall be in accordance with 3.6.1.

4.5.2.1.2 D.C. current gain.- With the test circuit in Figure 4 connected in succession to each of the circuits in Table XIII and under the test conditions noted in Table XIV, the current I_B shall be within the range shown in Table XIV, column "60° to 90°F", to determine compliance with 3.6.2.

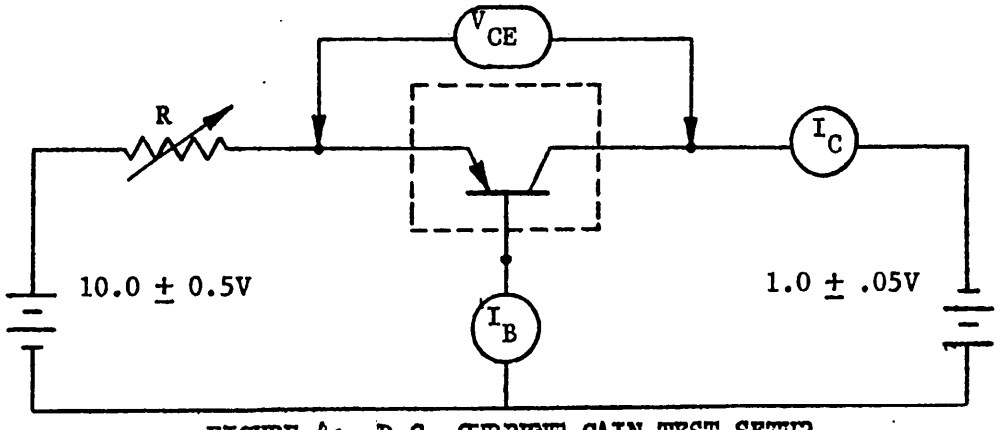


FIGURE 4: D.C. CURRENT GAIN TEST SETUP

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TABLE XIII

PIN IDENTIFICATION

Circuit No.	Emitter	Base	Collector
1	13A	10A	1B
2	1A	10B	18A
3	36A	33A	24B
4	24A	33B	40A
5	62A	59A	50B
6	50A	59B	45A

TABLE XIV

D.C. CURRENT GAIN TEST

	Test Conditions		Operating Temperatures			
			-40°F		60° to 90°F	
Circuit No.	V_{CE} (volts D.C.)	I_C (Amps D.C.)	I_B (ma. D.C.) Min.	Max.	I_B (ma. D.C.) Min.	Max.
1	-1.5 ± 0.5	3.0 ± 0.1	28.2	116.0	28.2	64.4
2	-1.5 ± 0.5	3.0 ± 0.1	28.2	116.0	28.2	64.4
3	-1.5 ± 0.5	3.0 ± 0.1	28.2	116.0	28.2	64.4
4	-1.5 ± 0.5	3.0 ± 0.1	28.2	116.0	28.2	64.4
5	-1.5 ± 0.5	3.0 ± 0.1	28.2	116.0	28.2	64.4
6	-1.5 ± 0.5	3.0 ± 0.1	28.2	116.0	28.2	64.4

4.5.2.1.3 D.C. Leakage current.- With the test circuit in Figure 5 connected in succession to each of the circuits in Table XIII and under the test conditions noted in Table XV, the leakage currents shall be less than the maximum values shown in the Table XV, column "60° to 90°F" to determine compliance with 3.6.2.

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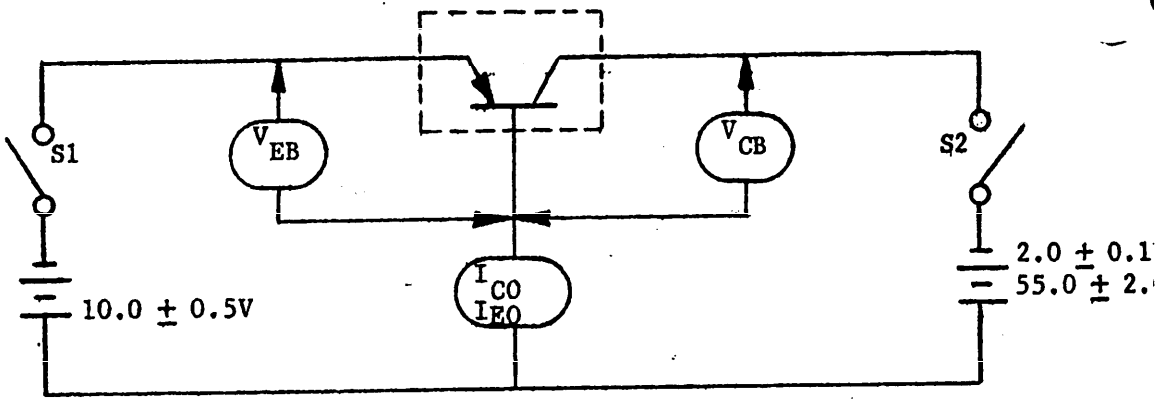


FIGURE 5: D. C. LEAKAGE CURRENT TEST SETUP

With S1 closed and S2 open, the ammeter reads I_{EO}

With S1 open and S2 closed, the ammeter reads I_{CO}

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TABLE XV

D.C. LEAKAGE CURRENT TEST

Circuit No.	Test Conditions S_1	Test Conditions S_2	Test Conditions Voltage (volts D.C.)	Operating Temperature 60° to 90°F I_{CO} & I_{EO} Max.	Operating Temperature +125°F I_{CO} & I_{EO} Max.
	open	closed	V_{CB} = -2.0 ± .2V	0.3 ma	--
1	open	closed	V_{CB} = -55 ± 3V	2.6 ma	6.0 ma
	closed	open	V_{EB} = -10 ± 1V	1.0 ma	--
	open	closed	V_{CB} = -2.0 ± .2V	0.3 ma	--
2	open	closed	V_{CB} = -55 ± 3V	2.6 ma	6.0 ma
	closed	open	V_{EB} = -10 ± 1V	1.0 ma	--
	open	closed	V_{CB} = -2.0 ± .2V	0.3 ma	--
3	open	closed	V_{CB} = -55 ± 3V	2.6 ma	6.0 ma
	closed	open	V_{EB} = .10 ± 1V	1.0 ma	--
	open	closed	V_{CB} = -2.0 ± .2V	0.3 ma	--
4	open	closed	V_{CB} = -55 ± 3V	2.6 ma	6.0 ma
	closed	open	V_{EB} = -10 ± 1V	1.0 ma	--
	open	closed	V_{CB} = -2.0 ± .2V	0.3 ma	--
5	open	closed	V_{CB} = -55 ± 3V	2.6 ma	6.0 ma
	closed	open	V_{EB} = -10 ± 1V	1.0 ma	--
	open	closed	V_{CB} = -2.0 ± .2V	0.3 ma	--
6	open	closed	V_{CB} = -55 ± 3V	2.6 ma	6.0 ma
	closed	open	V_{EB} = -10 ± 1V	1.0 ma	--

4.5.2.1.4 Operating temperature extremes.- The following tests shall be performed at the test conditions specified in 4.5.1.5 to determine conformance with 3.5.4 and 3.6.3.

4.5.2.1.4.1 D.C. current gain.- The test procedure of 4.5.2.1.2 is applicable except the current reading I_B shall be within range shown in the Table XIV, column "-40°F".

4.5.2.1.4.2 D.C. leakage current.- The test procedure of paragraph 4.5.2.1.3 is applicable except the leakage currents shall be less than the maximum values shown in the Table XV, column "+125°F".

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4.5.2.2 Amplifier, Power Supply: 10531686.- Each unit shall be tested in accordance with the test conditions specified herein to determine compliance with 3.7. Unless otherwise specified, electrical tests shall be performed at the standard ambient temperature (60° to 90°F).

4.5.2.2.1 Resistance.- Using an ohmmeter (Item 6, para. 4.4.1.1) measure the resistance between the pins of Table XVI for the values indicated. The proper polarity is noted. The DC resistance in ohms shall be in conformance to the requirements of 3.7.1.

TABLE XVI

ELECTRICAL RESISTANCE

Identity and Polarity of Pin Connector or Test Point		D.C. Resistance in OHMS		Req't Para.
(Positive)	(Negative)	(Min.)	(Max.)	
P-21	P-27	1,500	4,000	3.7.1(a)
P-25	P-22	1,500	4,000	3.7.1(a)
P-26	P-22	1,500	4,000	3.7.1(a)
P-27	P-22	1,500	4,000	3.7.1(a)
P-13	P-8	1,500	4,000	3.7.1(a)
P-12	P-8	1,500	4,000	3.7.1(a)
P-15	P-8	1,500	4,000	3.7.1(a)
P-19	P-13	1,500	4,000	3.7.1(a)
P-19	P-12	1,500	4,000	3.7.1(a)
P-19	P-15	1,500	4,000	3.7.1(a)
P-21	P-25	1,500	4,000	3.7.1(a)
P-21	P-26	1,500	4,000	3.7.1(a)
P-8	P-3	9,000	13,000	3.7.1(b)
P-43	P-39	9,500	11,000	3.7.1(c)
TP-28	P-48	9,500	11,000	3.7.1(c)
TP-28	P-51	9,500	11,000	3.7.1(c)
P-23	P-29	140	160	3.7.1(e)
P-57	P-65	25	29	3.7.1(f)
P-25	P-21	200K	---	3.7.1(g)
P-26	P-21	200K	---	3.7.1(g)
P-27	P-21	200K	---	3.7.1(g)
P-22	P-25	200K	---	3.7.1(g)
P-22	P-26	200K	---	3.7.1(g)
P-22	P-27	200K	---	3.7.1(g)

4.5.2.2.2 Reverse leakage of +25V bridge diodes.- To determine compliance with 3.7.2, apply -150 volts to the pins of column A, Table XVII through a 1 megohm, 1/4 watt, 5% resistor and ground the pins of column B. Measure a voltage of no greater than 0.3 volt across the resistor.

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TABLE XVII

Column A	Column B
Pin 19	Pin 13
Pin 19	Pin 12
Pin 19	Pin 15
Pin 15	Pin 8
Pin 12	Pin 8
Pin 13	Pin 8

4.5.2.2.3 Plus 6 volt supply.

(a) To determine compliance with 3.7.3, connect a 2N1542 transistor to the +6 supply as follows: base to pin 5, emitter to pin 3, and collector to pin 9. Connect a -20 volt supply to pin 1, +25 volts to pin 8, and ground at pin 22.

The voltage at pin 9 shall be +6 ± 0.5 volts.

(b) Connect an 80 ohms, 1 watt, 5% resistor to pin 9. The voltage at pin 9 shall be +6 ± 0.5 volts.

4.5.2.2.4 Plus 25 volt current limiter.

(a) To determine compliance with 3.7.4, make the connections of 4.5.2.2.3(a). The voltage at pin 2 shall be +24.0 ± 1.5 volts with respect to the +25 volt supply.

(b) Connect a 40 ohm, 10 watt, 5% resistor from -20V to pin 4. The voltage at pin 2 shall be between -0.1 volt and -1 volt with respect to ground.

(c) Remove all connections.

4.5.2.2.5 Minus 6 volt supply.- To determine compliance with 3.7.5, connect -20 volts to pin 1, and ground pin 22. The voltage at pin 24 shall be -6 ± 0.3 volts.

4.5.2.2.6 Minus 15 volts and minus 12.5 volt supplies.

(a) To determine compliance with 3.7.6, connect a 2N1542 transistor into this circuit as follows:

base to pin 30, emitter to pin 31, collector to -20V. Apply -20V to pin 1, and ground at pin 22. The voltage at pin 31 shall be -15 ± 1.5 volts. The voltage at pin 32 shall be -12.5 ± 1.5 volts.

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(b) Connect a 100 ohm, 2 watt, 5% resistor between pin 31 and ground, and a 200 ohm, 1/2 watt, 5% resistor between pin 32 and ground. The voltage at pin 31 shall be -15 ± 1.5 volts. The voltage at pin 32 shall be -12.5 ± 1.5 volts.

4.5.2.2.7 Minus 20 volt supply, current limiter, and voltage cut-out.- The following test shall be conducted to determine compliance with 3.7.7.

(a) Connect a 2N1542 transistor into the circuit as follows:

base to pin 11, emitter to pin 17, and collector to -30 volts. Connect -150 volts to pin 21, and ground at pin 22. Turn on the -30 volt supply before the -150 volts. The voltage at pin 1 shall be -19 ± 2 volts.

(b) Connect 10K ± 5% 25 watt resistor between -150V supply to pin 11. Connect a 36 ohm, 10 watt, 5% resistor to pin 1. The voltage at pin 1 shall be no more negative than -5 volts. The voltage at TP9 shall be between 5.7 volts and 6.8 volts with respect to pin 1.

(c) Remove the 36 ohm resistor. The voltage at pin 1 shall return to -19 ± 2 volts.

(d) Remove the -150 volt supply momentarily. After the -150 volts is reapplied the voltage at pin 1 shall be -19 ± 2 volts.

4.5.2.2.8 Start trigger; flip-flop, read indicator; power transistor driver, interlock switch; stop trigger.- The following tests to determine compliance with 3.7.8 shall be accomplished by making the following supply connections.

+ 6 volts to pin 38
- 15 volts to pin 46
ground to pin 22

(a) Apply -12.5 ± 0.6 volt to pin 35, and ground at pins 36 and 39. The following conditions shall exist within the circuit:

(1) The voltage at TP16 shall be between 0 and 0.5 volt.

(2) The voltage at TP17 shall be -9.0 ± 1.5 volts.

(3) The voltage at pin 54 with respect to -15 volts shall be less than 0.16 volt.

(4) The voltage at pin 62 shall be -7.1 ± 0.4 volts.

(5) The voltage at TP24 with respect to -15 volts shall be less than 0.16 volt.

(6) Reverse the voltages of pins 35 and 36 and measure pin 62 for -7.1 ± 0.4 volts.

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(b) Leave (a) unchanged and apply -12.5 volts to pin 51.

(1) The voltage at pin 62 shall be between 0 and -0.5 volt.

(2) The voltage at pin 54 shall be -15 ± 1.5 volts.

(3) The voltage at pin 64 with respect to +6 volts shall be between 5.2 volts and 5.3 volts.

(c) Remove -12.5 volts from pins 36 and 51. Ground both these pins. The following conditions shall exist within the circuitry.

(1) The voltage at TP16 shall be more negative than -8.4 volts.

(2) The voltage at TP17 shall be between 0 and -0.5 volt.

(3) The voltage at pin 54 shall be between 0 and -0.5 volt.

(4) The voltage at pin 62 shall be between 0 and -0.5 volt.

(5) The voltage at TP23 with respect to -15 volts shall be no greater than 0.2 volt.

4.5.2.2.9 Operating temperature extremes.- All tests specified in 4.5.2.2 shall be conducted at the temperatures specified in 3.5.4 to determine compliance with 3.7.9.

4.5.2.3 Electronic Control Amplifier: 10531690.- Each unit shall be tested in accordance with the test conditions specified herein to determine compliance with 3.8. Unless otherwise specified, electrical tests shall be performed at the standard ambient temperature (60° to 90°F).

4.5.2.3.1 Brake-solenoid circuit.

4.5.2.3.1.1 Solenoid diode.- The following procedures shall be applied to determine compliance with the requirements of 3.8.1.1.

(a) Forward drop: Apply -150 ± 5 volts through a 1K ohm, 30 watt, 5% resistor to pin 20, ground pin 26. The voltage at pin 20 shall be less than 1.0 volt.

(b) Reverse leakage: Apply -150 volts through the 1K ohm resistor to pin 26, and ground pin 20. The voltage across the 1K resistor shall be no more than 15 milli-volts.

4.5.2.3.1.2 Resistance-capacitance check.- An ohmmeter connected between pins 22 and 26 shall indicate a resistance of 51K ohms ± 5%. A capacitance checker between pins 26 (negative) and 25 shall indicate a value of 13 micro-farads, plus 3.0, minus 1.5 micro-farads in compliance with 3.8.1.2.

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4.5.2.3.2 Minus 3 volt supply.- The following procedures shall be utilized to determine compliance with 3.8.2:

(a) An ohmmeter connected between pins 35 and 60 shall indicate a resistance of 470 ohms ± 5%.

(b) Connect -4.5 ± 0.4 volts to pin 60. Ground pin 25. The voltage at pin 35 shall be -3.5 ± 0.5 volts.

4.5.2.3.3 Resistor check.- An ohmmeter connected between pins 17 and 25 shall indicate a resistance of 91K ohms ± 5% to determine compliance with 3.8.3.

4.5.2.3.4 Capacitance of AGC networks.- A capacitance checker shall be applied between pin 25 (positive) and each of pins 27, 29, 30, 31, 32, 33, 24, and 63 individually. Each reading shall indicate a capacitance of 1 microfarad ± 20%. The measurement between pin 25 and 61 shall be 15 ± 1.5 microfarads in compliance with the requirements of 3.8.4.

NOTE: For the following tests connect the power supplies as shown in Table XVIII

TABLE XVIII

-15 ± 0.7 volts	to	pin 2
+6 ± 0.3 volts	to	pin 10
-23 ± 1 volts	to	pin 58
ground	to	pin 25

4.5.2.3.5 Power transistor driver, solenoid switch.- Connect all supplies as per Table XVIII to determine compliance with 3.8.5

4.5.2.3.5.1 With pin 16 grounded, the voltage at pin 24 shall be -5.75 ± 0.4 volts.

4.5.2.3.5.2 With -15 ± 1 volts applied to pin 16,

(a) The voltage at TP18 shall be -7.4 ± 0.5 volts.

(b) The voltage at pin 24 shall be more positive than + 0.5 volt.

4.5.2.3.6 Inverter networks.- Connect all supplies as per Table XVIII to determine compliance with 3.8.6.

4.5.2.3.6.1 Ground pin 3 and apply -15 volts thru a 1K ohm, 5%, 1/4 watt resistor to pin 11. Read the following voltages for compliance with 3.8.6.1:

(a) Between -14.1 volts and -15.6 volts at pin 9.

(b) Between 0 and -0.5 volt at pin 13.

(c) Less than 0.1 volt across the 1K ohm test resistor.

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4.5.2.3.6.2 Interchange the voltages at pins 3 and 11. Read the following voltages for compliance with 3.8.6.2.:

(a) Between 0 and -0.5 volt at pin 9.

(b) Between -14.1 volts and -15.6 volts at pin 13.

(c) Less than 0.1 volt across the 1K ohm test resistor.

4.5.2.3.6.3 Ground both of pins 3 and 11. Read the following voltages for compliance with 3.8.6.3.:

(a) Between -14.1 volts and -15.6 volts at each of pins 9 and 13.

4.5.2.3.6.4 Ground pins 3 and 15, apply -15 volts thru a 1K ohm resistor to pin 8 to determine compliance with 3.8.6.4:

(a) The voltage across the 1K ohm resistor shall be less than 0.1 volt.

(b) Reverse the connections at pins 15 and 8, and repeat the measurement of (a).

4.5.2.3.7 Stop switch.- Connect all supplies as per Table XVIII and ground pin 3. Monitor pin 12 with an oscilloscope. As -15 volts is applied to pin 14, a pulse shall be produced on the scope consistent with Figure 1 to determine compliance with 3.8.7.

4.5.2.3.8 Delay and delay-inhibit.- Connect all supplies as per Table XVIII to determine compliance with 3.8.8. Ground pin 3. Monitor TP12 with an oscilloscope which is "synched" on pin 14. Apply -15 ± 0.7 volts to pin 14, and watch the scope for a response which is at least 80 milli-seconds but not greater than 400 milli-seconds from the input. The response voltage level shall start at -0.2V ± 0.02 V and after the delay shall go to a voltage no less negative than -13 volts.

4.5.2.3.9 Emitter followers:- Connect all supplies as per Table XVIII and the following, to determine compliance with 3.8.9:

(a) Ground pin 65. Connect a 10K ohm, 5%, 1/4 watt resistor between ground and pin 61. The voltage across the 10K ohm resistor shall be between 0 and 0.5 volt. Remove the input connections.

(b) Apply -15 ± .7V thru a 27K ohm, 5%, 1/4 watt resistor to pin 65. Connect a 300K ohm, 1/4 watt, 5% resistor between ground and each of pins 27, 29, 30, 31, 32, 33, 34, 63, and 61, one at a time. The voltage across the 300K ohm resistor shall be between -14.1V to -15.6V.

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4.5.2.3.10 Voltage conversion emitter followers.- Connect all supplies as per Table XVIII and the following to determine compliance with 3.8.10:

(a) Connect minus 3 volts to pin 36; connect -12.5 ± 0.7 volts to pins 38,40, 45, 47, 53, 56, 59, 62, and 64; connect a 150 ohm, 5%, 1/2 watt resistor to each of pins 37, 42, 43, 49, 51, 52, 54, 55, and 57. The voltage across the 150 ohm resistor shall be -2.7 ± 0.3 volts.

(b) Replace the -12.5 volt inputs with ground. The voltage across the 150 ohm resistor shall be between 0 and -0.5 volt.

4.5.2.3.11 Temperature extremes.- The following tests shall be performed at the test conditions of 4.5.1.5 to determine compliance with 3.8.11.

4.5.2.3.11.1 Brake-solenoid circuit.- Repeat 4.5.2.3.1.1(a) and (b), except as follows, to determine compliance with 3.8.11.1:

(a) The voltage specified in (a) shall be less than 1.5 volts.

(b) The voltage specified in (b) shall be no more than 50 millivolts.

4.5.2.3.11.2 Minus 3 volt supply.- Repeat 4.5.2.3.2(b) except the voltage at pin 35 shall be -3.5 ± 0.3 volts to determine compliance with 3.8.11.2.

4.5.2.3.11.3 Power transistor drives.- Repeat 4.5.2.3.5 except as follows to determine compliance with 3.8.11.3.

(a) At condition 1, the voltage at pin 24 shall be -5.7 ± 0.5 volts.

(b) At condition 2a, the voltage at TP18 shall be -7.4 ± 0.7 volts.

(c) At condition 2b, the voltage at pin 24 shall be more positive than ± 0.3 volt.

4.5.2.3.11.4 Inverter networks.- Repeat 4.5.2.3.6 except in 1c, 2c, and 4c, the voltage across the 1K ohm resistor shall be less than 0.3 volt to determine compliance with 3.8.11.4.

4.5.2.3.11.5 Stop switch.- Repeat 4.5.2.3.7 for compliance with 3.8.11.5.

4.5.2.3.11.6 Delay, delay-inhibit.- Repeat 4.5.2.3.8 for compliance with 3.8.11.6.

4.5.2.3.11.7 Emitter follower.- Repeat 4.5.2.3.9 for compliance with 3.8.11

4.5.2.3.11.8 Voltage conversion emitter followers.- Repeat 4.5.2.3.10 except for the following to determine compliance with 3.8.11.8.

(a) The voltage across the 150 ohm resistor in (a) shall be -2.7 ± 0.5 volts.

(b) The voltage across the resistor in (b) shall be between 0 and -0.6 volt.

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4.5.2.4 Photodiode Amplifier: 8213319.- Each unit shall be tested in accordance with the test conditions specified herein to determine compliance with 3.9. Unless otherwise specified, electrical tests shall be performed at the standard ambient temperature (60° to 90°F).

4.5.2.4.1 For the collowing tests connect all supplies as follows:

-12.5	±	0.2 volts	to	pin 9
+6	±	0.1 volts	to	pin 20
-15	±	0.2 volts	to	pin 8
ground			to	pins 7, 11, 18, 26, 37, 43, and 56

4.5.2.4.2 Resistor selection.- To determine compliance with 3.9.1, connect a 50K ohm, 1/4 watt, 10 turn, potentiometer in series with a 60K ohm, 1/4 watt resistor. Place this combination in the mounting terminals of the missing component slots R-56 thru R-63 one at a time. Connect -15 volts to pin 10. With a signal generator (Item 9) supplying the input of Figure 2A to the pins of column A, Table XIX vary the pot until the output at the corresponding pins of column B, Table XIX is as defined in Figure 2B. Remove the resistor-pot combination and measure the total resistance on a resistance bridge (Item 10). Select a fixed resistor from those available which has a value within 500 ohms of the measured resistance.

After installation of the resistors, reapply the input wave and check the output for the output of Figure 2B.

TABLE XIX

Column A	Column B
Pin 15	Pin 13
Pin 34	Pin 32
Pin 47	Pin 49
Pin 50	Pin 62
Pin 24	Pin 14
Pin 31	Pin 33
Pin 48	Pin 53
Pin 65	Pin 63

4.5.2.4.3 Photodiode amplifier inhibit diodes.- With the setup used in 4.5.2.4.4 and the Figure 2B output, application of ground to pin 10 shall cause each output to hold a +6 ± 0.5 volt level to determine compliance with 3.9.2.

4.5.2.4.4 TR amplifier.- Connect all supplies as in 4.5.2.4.1 and the following to determine compliance with 3.9.3.

4.5.2.4.4.1 Connect -15 volts to pin 10.

(a) The voltage at TP45 shall be -6.2 ± 0.4 volts.

(b) The voltage at pin 5 shall be -12.3 ± 0.7 volts.

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4.5.2.4.4.2 Connect ground to pin 10. The voltage at pin 5 shall be +6 ± 0.3 volts.

4.5.2.4.5 Clock amplifier.- Connect all supplies as per 4.5.2.4.1 and the following, to determine compliance with 3.9.4. Connect -15 volts through 3.9K ± 5% resistor to pin 1. Connect a square wave input similar to Figure 2A except the peak-to-peak value is 4 ± 0.2 volts, and the frequency is 488 ± 20 cycles per second to pin 4. Connect an oscilloscope to pin 3. The output shall be as shown in Figure 3. Connect the scope to pin 1. The output shall be as shown in Figure 3 except the negative limit shall be -12.5 ± 1.5 volts.

4.5.2.4.6 Operating temperature extremes.- The following tests shall be performed at the test conditions of 4.5.1.5 to determine compliance with 3.9.5.

4.5.2.4.6.1 Photodiode amplifier and inhibit diode.- Connect all supplies as per 4.5.2.4.1. Connect -15 volts to pin 10. Connect the input of Figure 2A to the pins of column A, Table XIX. An oscilloscope connected to the corresponding pins of column B, Table XIX shall show a display with the applicable readings of Figure 2B in compliance with 3.9.5.1. Apply ground to pin 10. The scope at the output pins shall show the output holding at +6 ± 0.6 volts.

4.5.2.4.6.2 TR amplifier.- Repeat 4.5.2.4.4 with the following changes to determine compliance with 3.9.5.2.:

(a) (1a) The voltage at TP45 shall be -6.2 ± 0.6 volts.

(b) (1b) The voltage at pin 5 shall be -12.3 ± 1 volts.

(c) (2) The voltage at pin 5 shall be +6 ± 0.6 volts.

4.5.2.4.6.3 Clock amplifier.- Repeat 4.5.2.4.5 to determine compliance with 3.9.5.3.

4.5.2.5 Signal Data Reproducer Sub-Assembly: 10531698.- Unless otherwise specified, the requirements of 3.10 shall be determined by the following tests conducted at the standard room temperature (60° to 90°F).

4.5.2.5.1 Wiring.- Wiring and connections shall be visually examined to Electrical Schematic Drawing F10531712 to determine compliance with 3.10.1.

4.5.2.5.2 Zener voltages.- To determine compliance with 3.10.2, apply -30 volts through a 14 ohm -20 watt, 5% resistor to J-9, P-9. Place ground at J-9, P-10. Read (-15.2) to (-16.8) volts at J-9, P-9.

4.5.2.5.3 Break solenoid switch.- Current and voltage readings shall be in compliance with 3.10.3 when tested as follows:

(a) Apply -2 volts directly to J-3, P-17. Place ground at J-3, P-24 through a microammeter (Item 4, paragraph 4.4.2.2). Read a maximum current of 100 micro-amps.

(b) Apply (-20) volts, through a 5ohm, 45 watt, 5% resistor to J-3, P-17. Apply (-20) volts, through a 400 ohm, 2 watt, 5% resistor to J-3, P-[illegible]. Place ground at J-2, P-22. The voltage across the 5 ohm resistor shall not less than 5.36 volts nor greater than 14.5 volts.

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4.5.2.5.4 Drive solenoid switch.- Micro-amp and voltage readings shall be in compliance with 3.10.4 when tested as follows:

(a) Apply -2 volts directly to J-2, P-65. Place ground at J-2, P-64 through a microammeter (Item 4 of 4.4.1.1). Read a maximum of 100 micro-amps.

(b) Apply (-20) volts through a 5 ohm, 45 watt, 5% resistor to J-2, P-65. Apply (-20) volts, through a 400 ohm, 2 watt, 5% resistor to J-2, P-64. Place ground at J-2, P-22. The voltage across the 5 ohm resistor shall not be less than 4.31 volts no greater than 11.6 volts.

4.5.2.5.5 Transformer supply.- Apply 115 ± 5 volts RMS (line to neutral), 3 phase, 400 ± 20 cycles to the transformer as follows to determine compliance with 3.10.5.

P-5 Phase "A"
P-3 Phase "B"
P-1 Phase "C"

Read the following A.C. voltages:

Between P-9 and P-8	read	23.8V RMS	± 5%
Between P-9 and P-7	read	23.8V RMS	± 5%
Between P-8 and P-7	read	23.8V RMS	± 5%
Between P-13 and P-14	read	110V RMS	± 5%
Between P-13 and P-15	read	110V RMS	± 5%
Between P-15 and P-14	read	110V RMS	± 5%
Between P-10 and P-11	read	26.5V RMS	± 5%
Between P-10 and P-12	read	26.5V RMS	± 5%
Between P-12 and P-11	read	26.5V RMS	± 5%

4.5.2.5.6 D.C. resistance of selected circuits.- Using an ohmmeter (Item 6 of 4.4.1.1) measure the resistance between the pins of Table XX for the values indicated. The proper polarity is indicated. Resistance and reverse impedance in OHMS shall be within the tolerance specified to insure compliance with 3.10.6.

TABLE XX

ELECTRICAL RESISTANCE

Identity and Polarity of Pin Connector		D.C. Resistance in OHMS and Tolerance
(Positive)	(Negative)	
J-1, P-1B	J-3, P-60	450 ± 3%
J-1, P-2B	J-2, P-34	39 ± 5%
J-1, P-4B	J-3, P-20	27 ± 5%
J-1, P-2B	J-9, P-9	25 ± 3%
J-1, P-11A	J-2, P-21	10,000 ± 3%
J-1, P-1A	J-2, P-4	4.7 ± 5%
J-1, P-36B	J-2, P-4	4.7 ± 5%
P-8 (transformer)	J-1, P-1B	50,000 or greater
P-16(transformer)	P-8 (transformer)	50,000 or greater

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4.5.2.5.7 Clock inhibit diode.- To determine compliance with 3.10.7, apply (-20) volts through 3,600 ohms, 1/2 watt, 5% resistor to J-4, P-2. Apply ground to J-3, P-15. The voltage with respect to ground, at J-4, P-2 shall not exceed (-0.75) volt. Apply (-20) volts, through a 10,000 ohm, 1/2 watt, 5% resistor to J-3, P-15. Place ground at J-4, P-2. The voltage across the 10,000 ohm resistor shall not exceed 0.525 volt.

4.5.2.5.8 Operating temperature extremes.- The following tests shall be performed at the test conditions specified in 4.5.1.5 to determine compliance with 3.10.8.

4.5.2.5.8.1 Zener voltages.- The test procedure of 4.5.2.5.2 is applicable for determining compliance with 3.10.8.1 except the measured voltages shall conform to the tolerances specified in 3.10.8.1 for CR8.

4.5.2.5.8.2 Brake solenoid switch.- To determine base collector cut-off current in compliance with 3.10.8.2 apply -100 volts to J-3, P-17. Place ground at J-3, P-24 through an ammeter (Item 8 of 4.4.1.1). Read a maximum current of 15 milliamps.

4.5.2.5.8.3 Drive solenoid switch.- To determine base collector cut-off current in compliance with 3.10.8.3 apply (-60) volts to J-2, P-65. Place ground at J-2, P-64 through an ammeter (Item 8 of 4.4.1.1). Read a maximum of 15 milliamps.

4.5.2.5.8.4 Transformer supply.- The test procedure of 4.5.2.5.5 is applicable for determining compliance with 3.10.8.4 except the measured A.C. voltage shall have a tolerance of ± 10%.

4.5.2.5.8.5 D.C. resistance of selected circuits.- The test procedure of 4.5.2.5.6 is applicable for determining compliance with 3.10.8.5.

4.5.2.5.8.6 Clock inhibit diode.- The test procedure of 4.5.2.5.7 is applicable for determining compliance with 3.10.8.6 except that the voltage read at J-4, P-2 shall not exceed 0.5 volt and the voltage read across the 10,000 ohm resistor shall not exceed 1.05 volts.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking.- Preservation, packaging packing and marking shall be in accordance with the following:

Transistor Assembly, Power: 10531682	Master Packaging Sheet 14
Amplifier Power Supply: 10531686	Master Packaging Sheet 14
Amplifier, Electronic Control: 10531690	Master Packaging Sheet 14
Amplifier Assembly, Photodiode: 8213319	Master Packaging Sheet 14
Signal Data Reproducer Sub-Assembly: 10531698	Packaging Data Sheet MIL-P-14232/P10531698

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the applicable drawings.

6. NOTES

6.1 Intended use.- Modules procured under this specification are for the production of the Reproducer, Signal Data: AN/GSQ-64 and as spare replacements for existing modules.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of specification
(b) Selection of an applicable level of preservation, packaging and packing
(c) Identity of module(s) to be procured
(d) Applicable stock number
(e) A block of serial numbers to identify each module

Custodian:

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Preparing activity:

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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